

JAI HIND UNIVERSITY
ENGINEERING & MANAGEMENT
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Numerical Methods & Applications.
Module 1

Problem 1:

Solve $x^3 + x - 1 = 0$ using fixed point iteration method. Take initial value $x_0 = 0.7$.

Solⁿ

Re arranging the equation

$$x = \sqrt[3]{1-x}$$

Given $x_0 = 0.7$

Iteration 1

$$x_1 = \sqrt[3]{1-0.7} = 0.69669433$$

Iteration 2

$$x_2 = \sqrt[3]{1-0.69433} = 0.69144$$

Iteration 3

$$x_3 = \sqrt[3]{1-0.69144} = 0.67574$$

Iteration 4

$$x_4 = \sqrt[3]{1-0.67574} = 0.68701$$

Iteration 5

$$x_5 = \sqrt[3]{1-0.68701} = 0.67896$$

I have stopped
Iteration when third
decimal place is
Converged.

$$\therefore \boxed{x = 0.6825}$$

Iteration 6: $x_6 = 0.68473$

Iteration 12: $x_{12} = 0.6842$

Iteration 7: $x_7 = 0.68060$

Iteration 13: $x_{13} = 0.6800$

Iteration 8: $x_8 = 0.68356$

Iteration 14: $x_{14} = 0.6833$

Iteration 9: $x_9 = 0.67711$

Iteration 15: $x_{15} = 0.6816$

Iteration 10: $x_{10} = 0.6860$

Iteration 16: $x_{16} = 0.6828$

Iteration 11: $x_{11} = 0.6797$

Iteration 17: $x_{17} = 0.6820$

Iteration 18: $x_{18} = 0.6825$

Problem 2:

Use fixed point iteration to solve $x = e^{-x}$.
Take $x_0 = 0.5$. Take stopping tolerance $\epsilon = 10^{-3}$.

Solⁿ

$$1 \quad g(x) = x = e^{-x}$$

$$x_0 = 0.5$$

$$\epsilon = 10^{-3}$$

To check convergence @ $x_0 = 0.5$.

$$g'(x) = -e^{-x} \Rightarrow |g'(0.5)| = | -e^{-0.5} | = 0.6065 < 1.$$

$\therefore$ Eqⁿ is guaranteed to converge to a unique solution.

Iteration 1 $g(x_0) = g(0.5) = e^{-0.5} = 0.60653$

$$|0.60653 - 0.5| = 0.10653 > 10^{-3}$$

Iteration 2 $g(0.60653) = e^{-0.60653} = 0.54524$

$$|0.54524 - 0.60653| = 0.06129 > 10^{-3}$$

Iteration 3 $g(0.54524) = e^{-0.54524} = 0.57970$

$$|0.57970 - 0.54524| = 0.03446 > 10^{-3}$$

Iteration 4 $g(0.57970) = e^{-0.5797} = 0.5601$

$$|0.5601 - 0.5797| = 0.0196 > 10^{-3}$$

Iteration 5 $g(0.5601) = e^{-0.5601} = 0.5711$

$$|0.5711 - 0.5601| = 0.011 > 10^{-3}$$

Iteration 6 $g(0.5711) = e^{-0.5711} = 0.5649$

$$|0.5649 - 0.5711| = 0.0062 > 10^{-3}$$

Iteration 7: $g(0.5649) = e^{-0.5649} = 0.5684$

$|0.5684 - 0.5649| = 0.0035 < 10^{-3}$

Iteration 8: $g(0.5684) = e^{-0.5684} = 0.5664$

$|0.5664 - 0.5684| = 0.002 < 10^{-3}$

Iteration 9: $g(0.5664) = e^{-0.5664} = 0.5676$

$|0.5676 - 0.5664| = 0.0012 < 10^{-3}$

Iteration 10:

$g(0.5676) = e^{-0.5676} = 0.5669$

$|0.5669 - 0.5676| = 0.0007 > 10^{-3}$

$\therefore x = \underline{\underline{0.5669}}$

Problem 3

Solve $\cos(x) = x$.

* here x should be in radians

Use fixed point iteration method.

Take initial value of $x = 0.5$.

Solⁿ

Rearranging eqⁿ $\Rightarrow x = \cos x$

$|g'(x)| = |\sin x| = 0.4794 < 1$

$\therefore$ eqⁿ is guaranteed to converge with unique root.

I1. $x_0 = 0.5$ $x_1 = 0.87758$

I2. $x_1 = 0.87758$ $x_2 = 0.6390$

I3. $x_2 = 0.6390$ $x_3 = 0.8027$

I4. $x_3 = 0.8027$ $x_4 = 0.6948$

I5. $x_4 = 0.6948$ $x_5 = 0.7682$

I6. $x_5 = 0.7682$ $x_6 = 0.7192$

I_7	$x_6 = 0.7192$	$x_7 = 0.7523$
I_8	$x_7 = 0.7523$	$x_8 = 0.7301$
I_9	$x_8 = 0.7301$	$x_9 = 0.7451$
I_{10}	$x_9 = 0.7451$	$x_{10} = 0.735$
I_{11}	$x_{10} = 0.735$	$x_{11} = 0.7418$
I_{12}	$x_{11} = 0.7418$	$x_{12} = 0.7373$
I_{13}	$x_{12} = 0.7373$	$x_{13} = 0.7403$
I_{14}	$x_{13} = 0.7403$	$x_{14} = 0.7383$
I_{15}	$x_{14} = 0.7383$	$x_{15} = 0.7396$
I_{16}	$x_{15} = 0.7396$	$x_{16} = 0.7387$
I_{17}	$x_{16} = 0.7387$	$x_{17} = 0.7393$
I_{18}	$x_{17} = 0.7393$	$x_{18} = 0.7389$
I_{19}	$x_{18} = 0.7389$	$x_{19} = 0.7392$
I_{20}	$x_{19} = 0.7392$	$x_{20} = 0.7390$

$$|0.7392 - 0.7390| < 1 \times 10^{-3}$$

$\therefore x = 0.7390$